Tecnología de los agregados

Rocas ígneas

Parte 4

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01

Texturas

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Texturas

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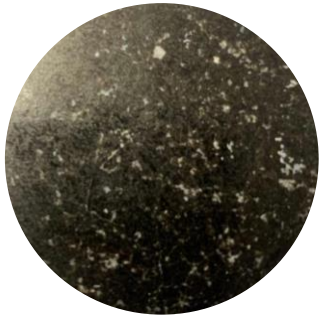
Texturas

Propiedades

- Tamaño

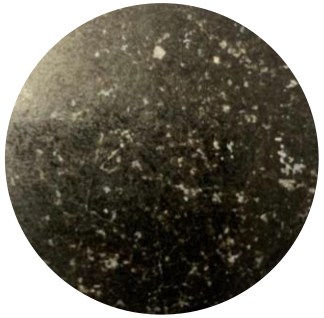


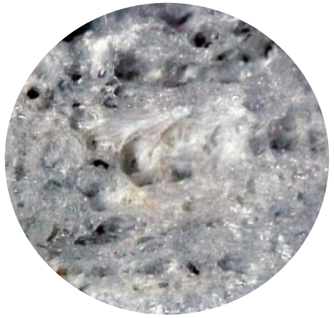
Texturas

Propiedades

- Tamaño
- Cristalinidad



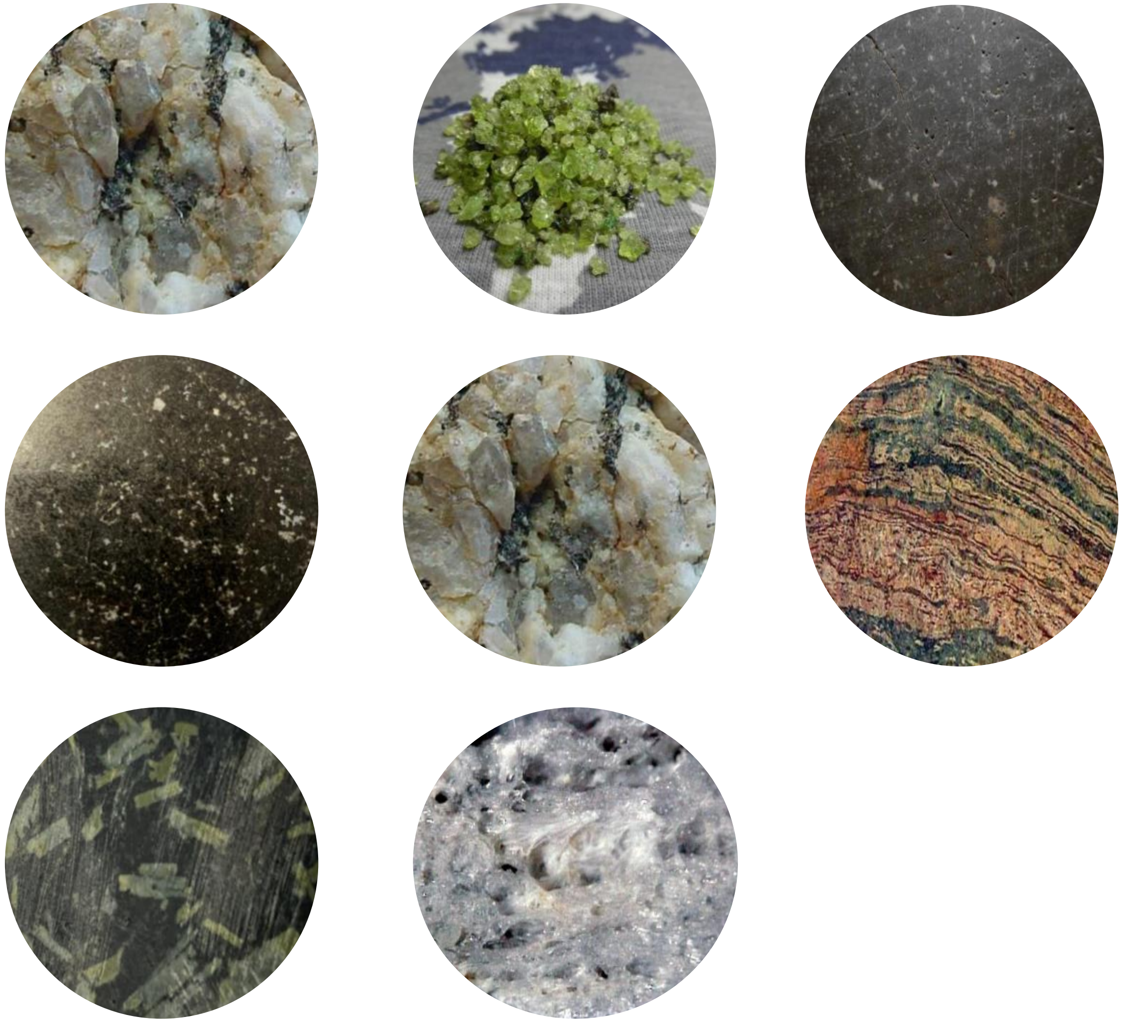


Texturas

Propiedades

- Tamaño
- Cristalinidad
- Fábrica



Texturas

- **Afanítica:** es cuando los cristales solo se pueden ver con ayuda del microscopio, menores de 0.5 mm.
- **Porfídica:** es una matriz de cristales microscópicos coexisten cristales observables a simple vista o con lupa.
- **Vesicular:** cuando la roca presenta pequeñas cavidades o espacios vacíos producto de la liberación de gases.
- **Amigdaloide:** cuando las cavidades de la roca están ocupados por mineral de relleno.
- **Piroclástica:** cuando las rocas se componen de fragmentos volcánicos (ceniza) que el depositarse se cementan o compactan.
- **Vítrea:** son rocas con vidrio o vidrio exclusivamente como la riolita con obsidiana, perlita, etc.

Texturas

Porfídica

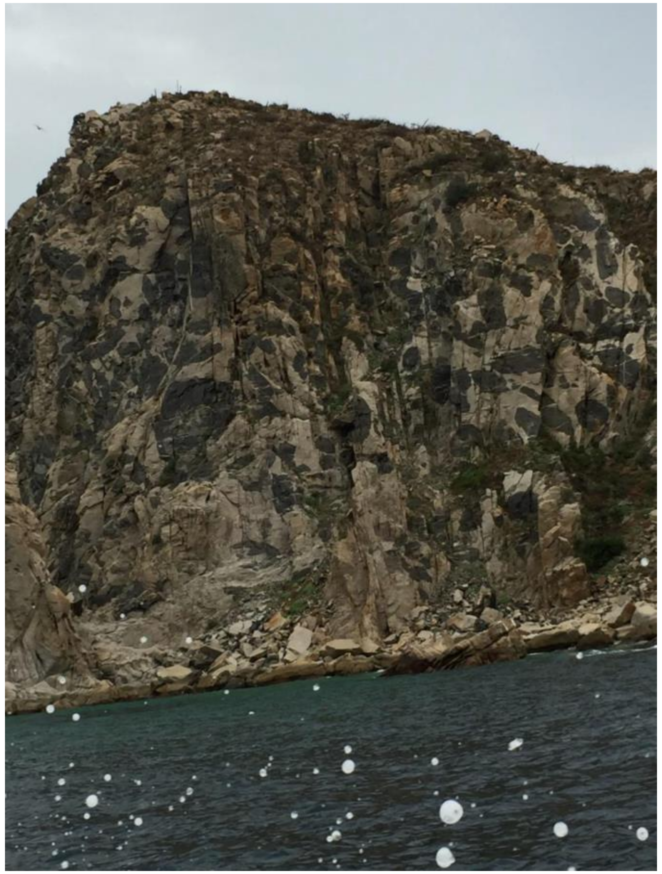
Esta textura es característica de las rocas de profundidad media (hipobisales), aunque también presentan rocas extrusivas.

Ejemplo:

En rocas extrusivas - Andesita porfídica

En rocas hipoabisales - Pórfido andesítico

Texturas Xenolitos



02

La Tierra

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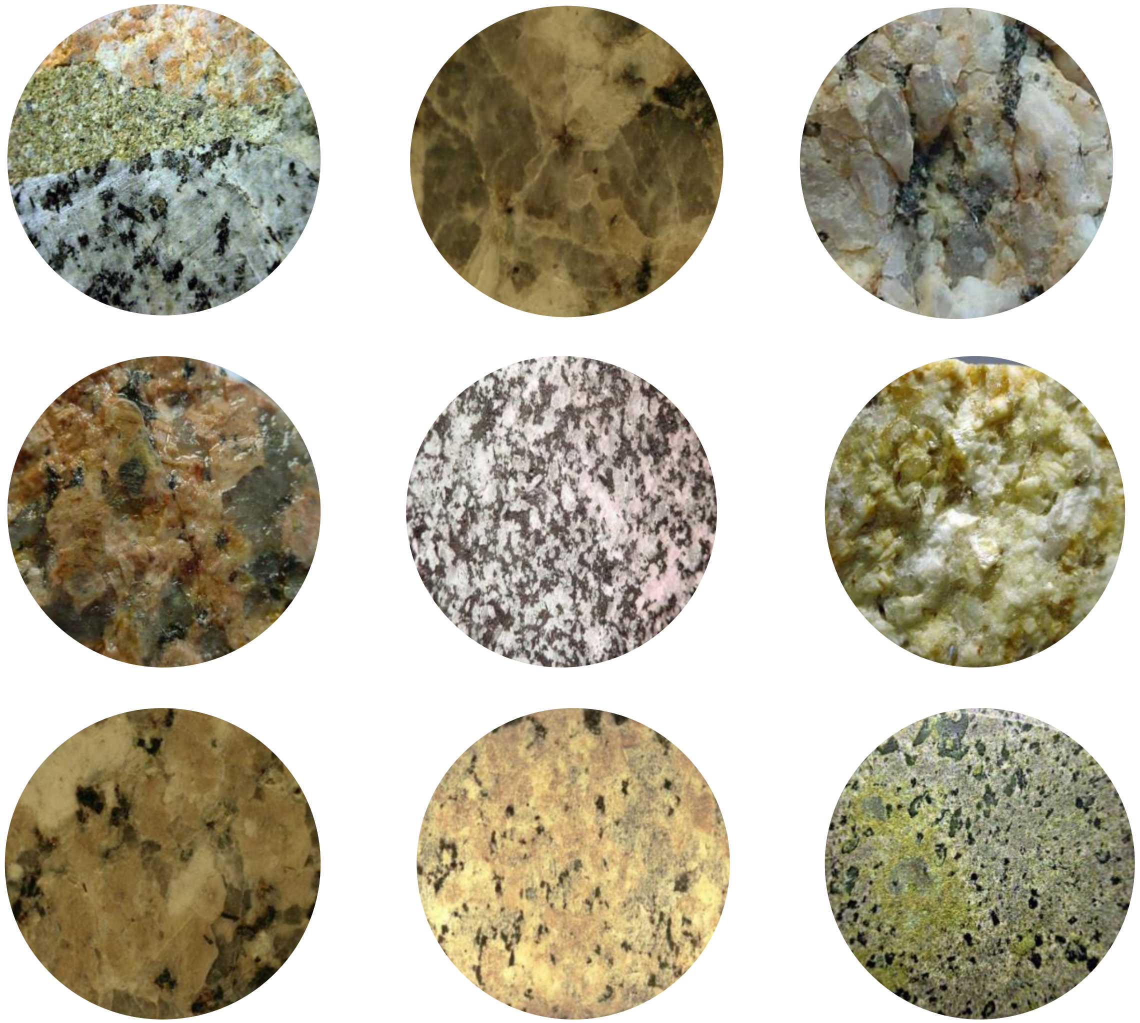
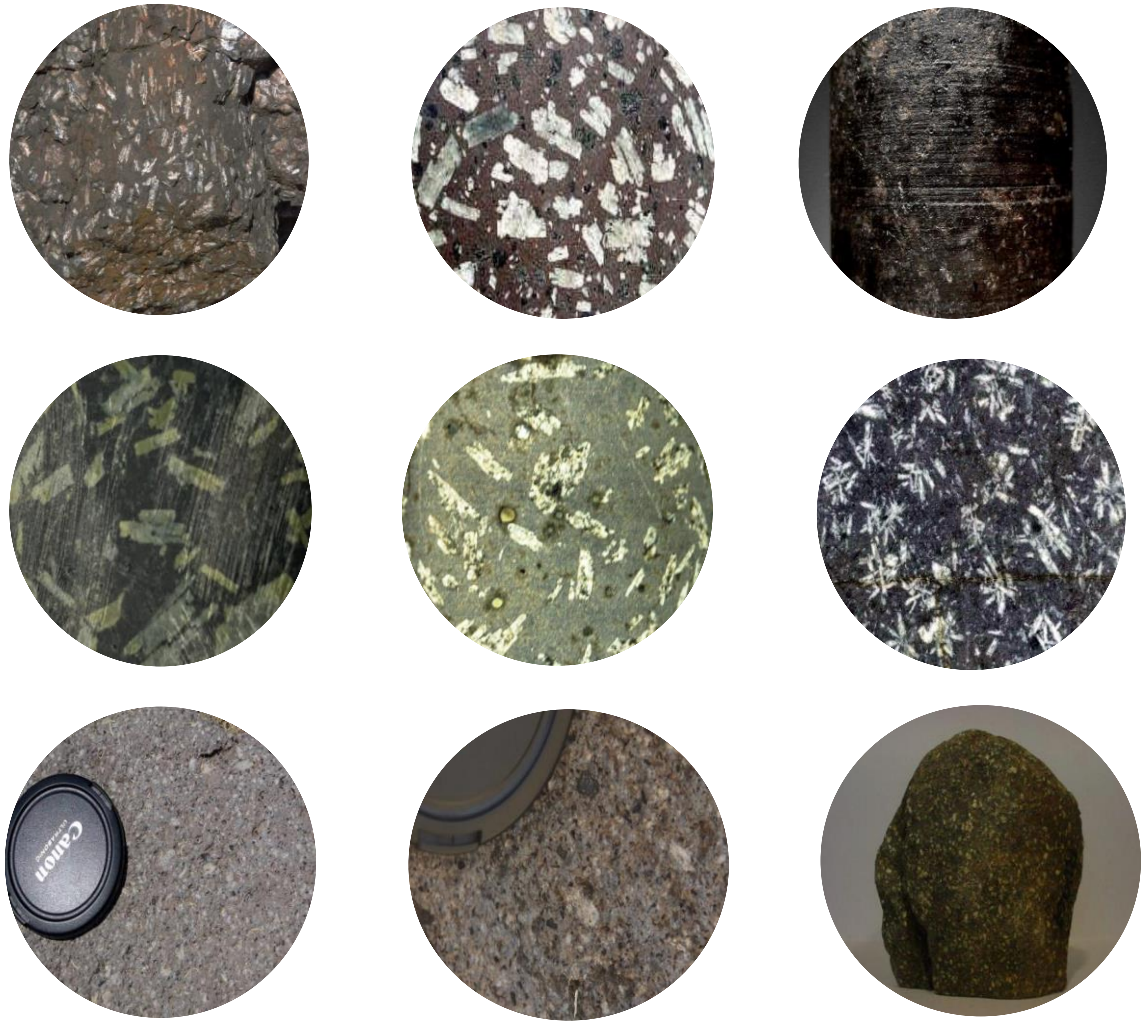
La Tierra



La Tierra



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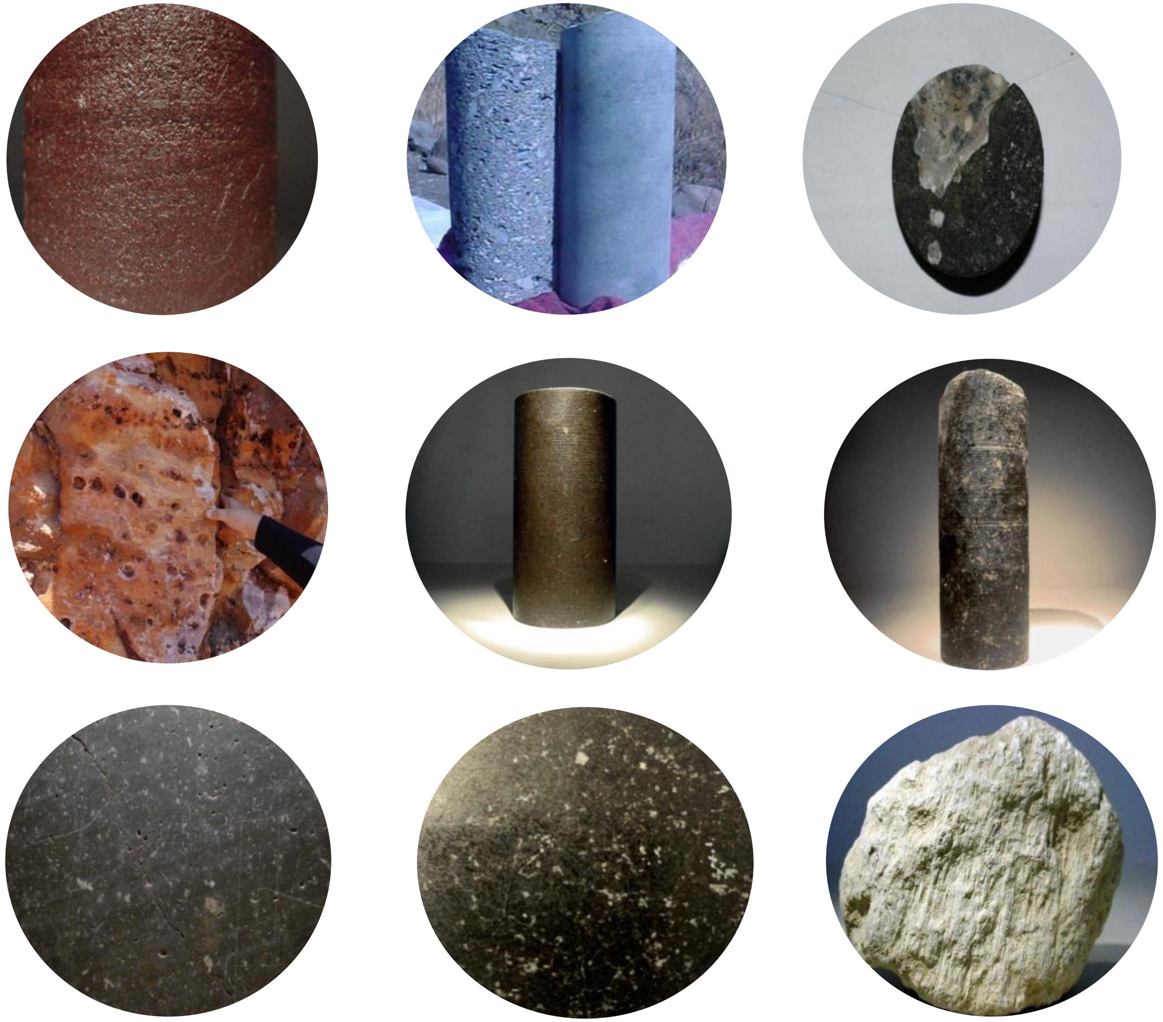

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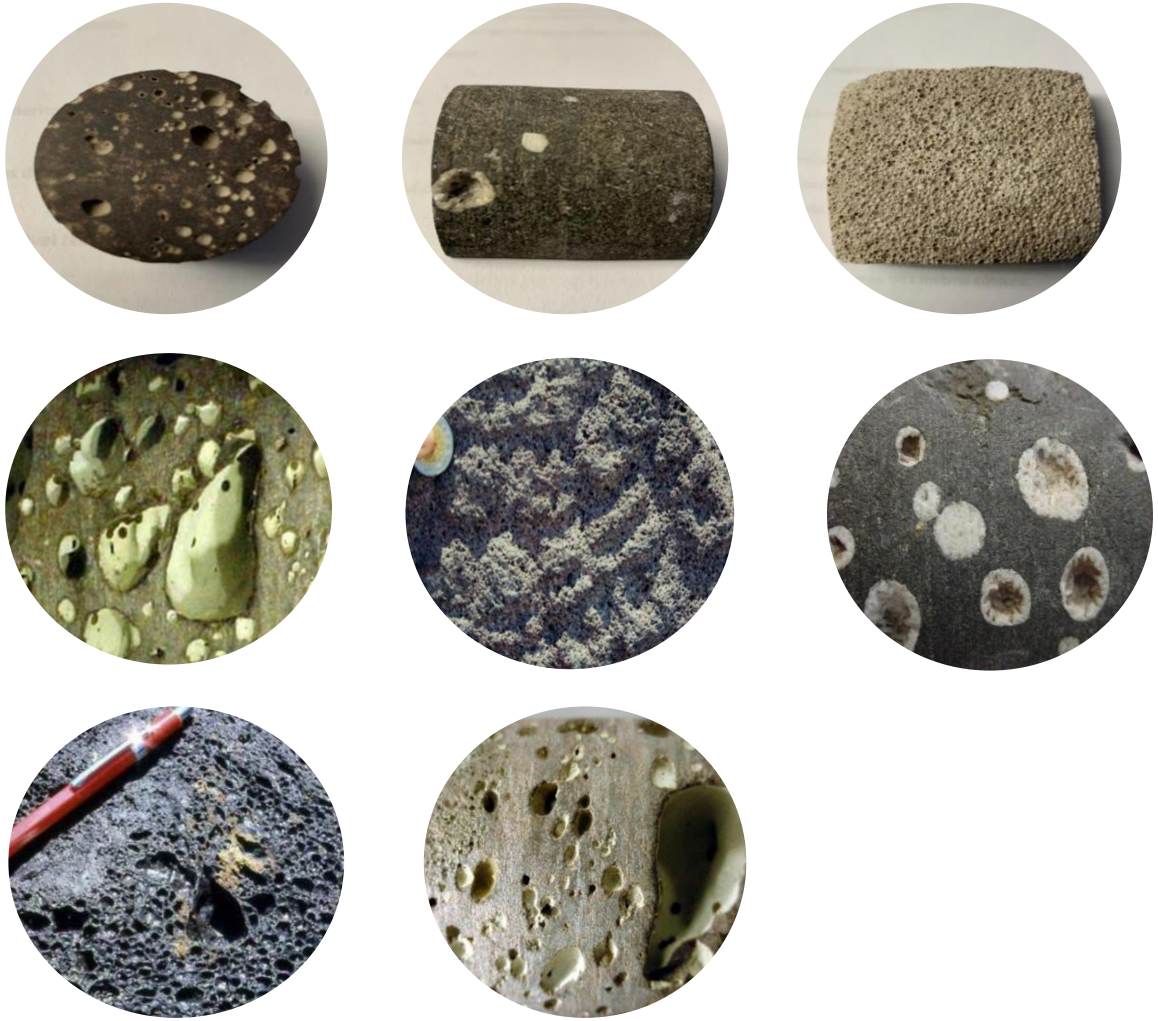
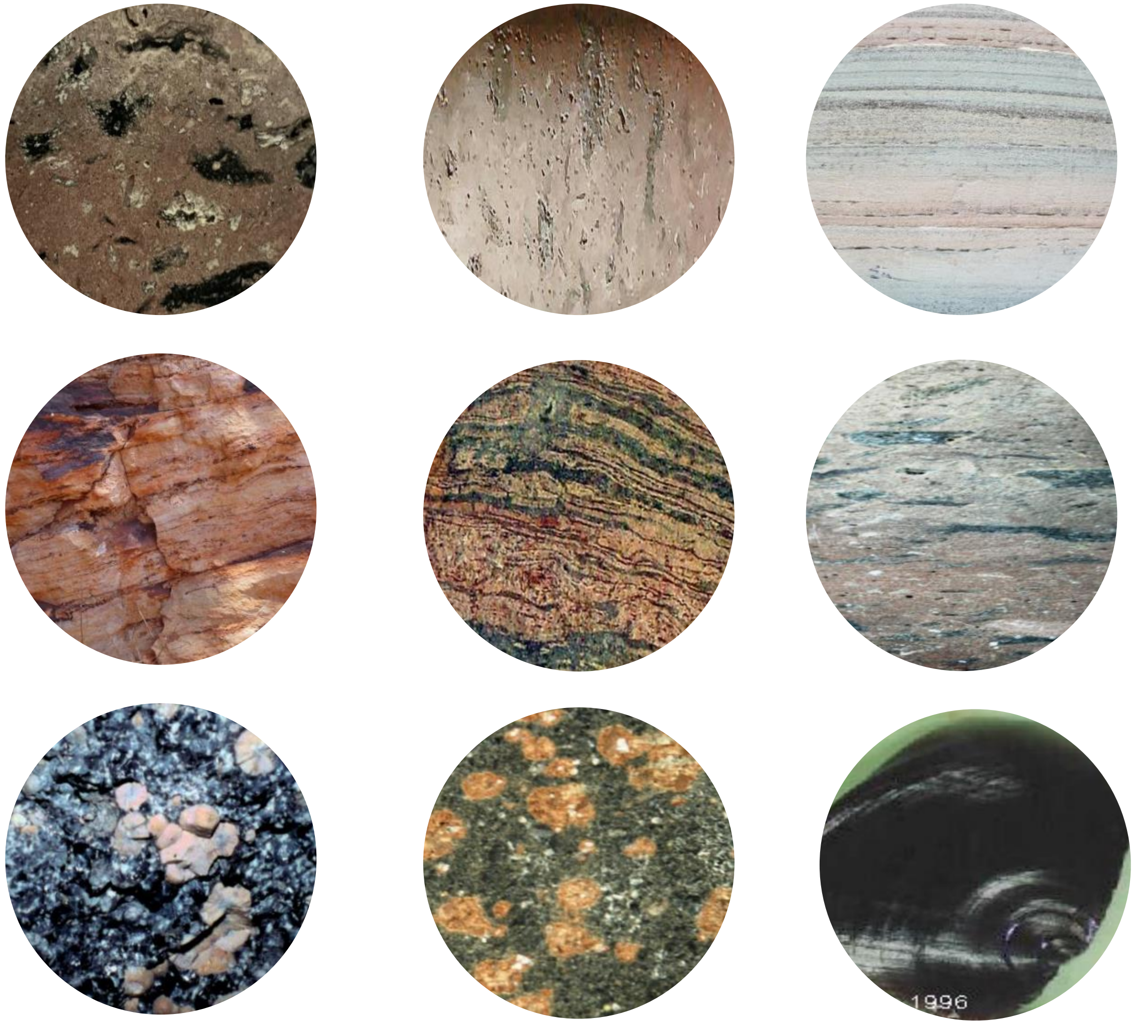


La Tierra



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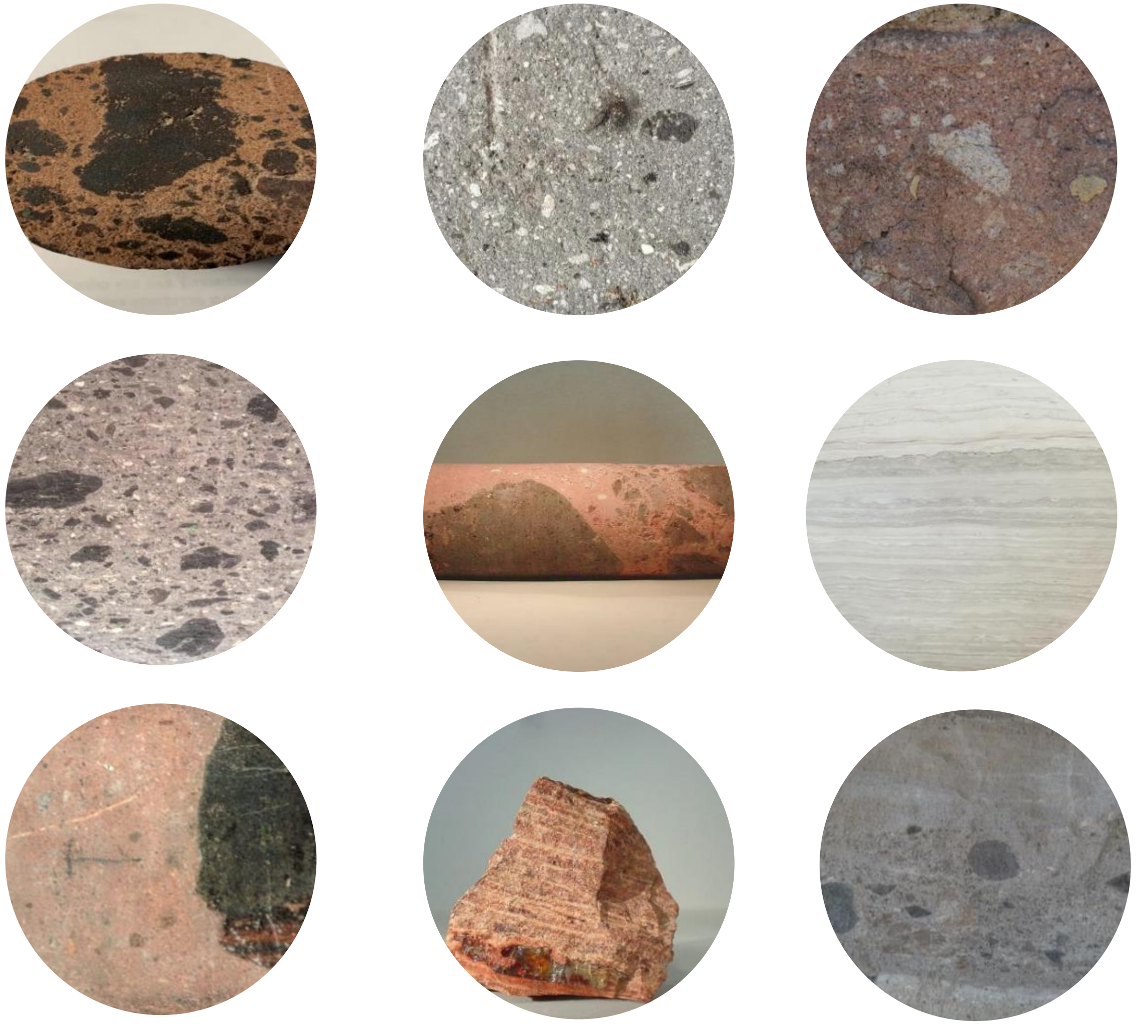
La Tierra

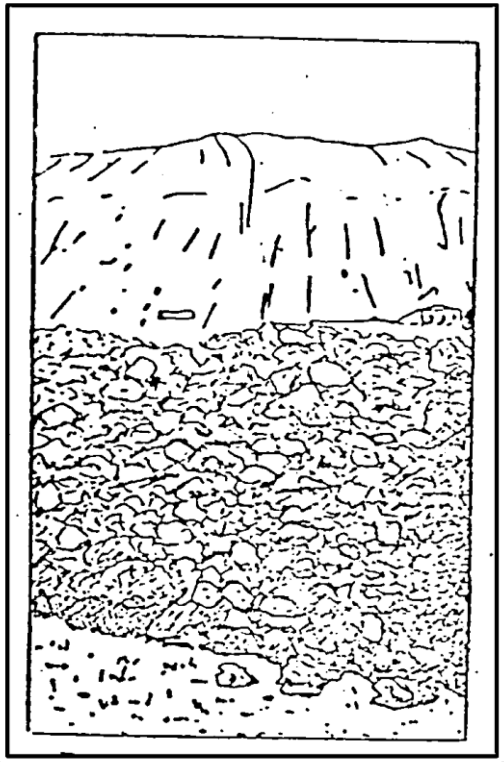


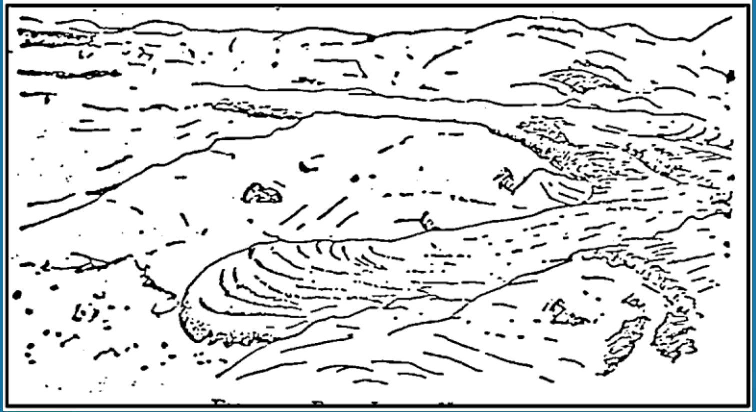
03

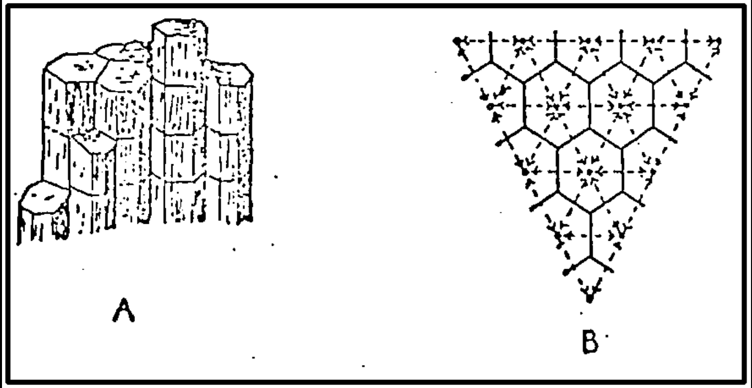
Estructuras

Estructura de las rocas ígneas

Estructuras







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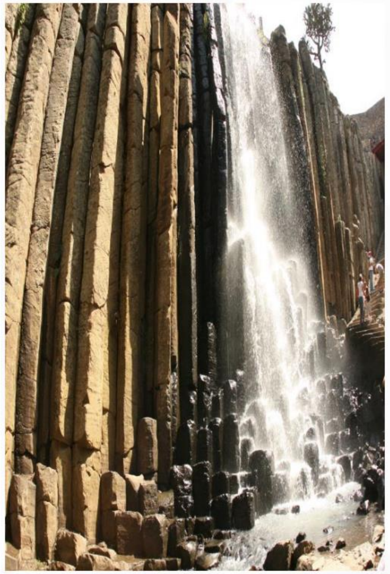
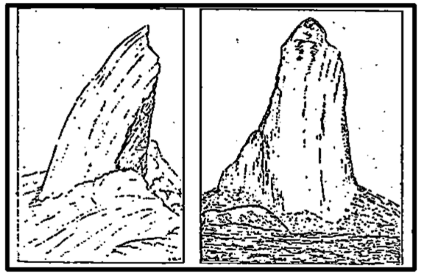
Estructuras

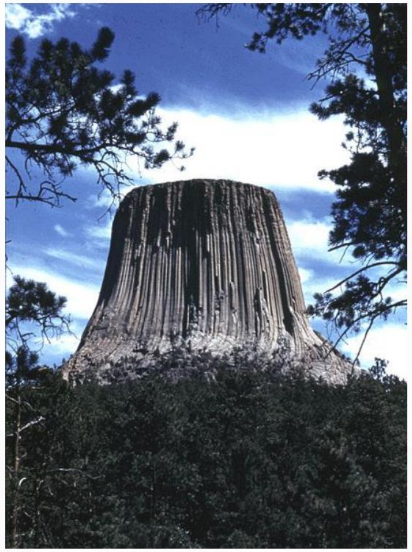


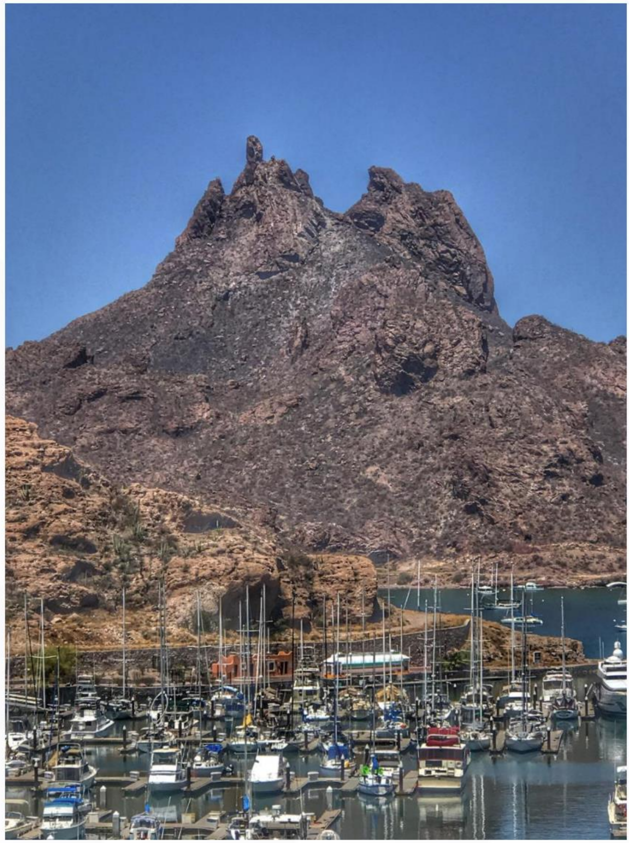




Estructuras







Estructuras

Estructuras

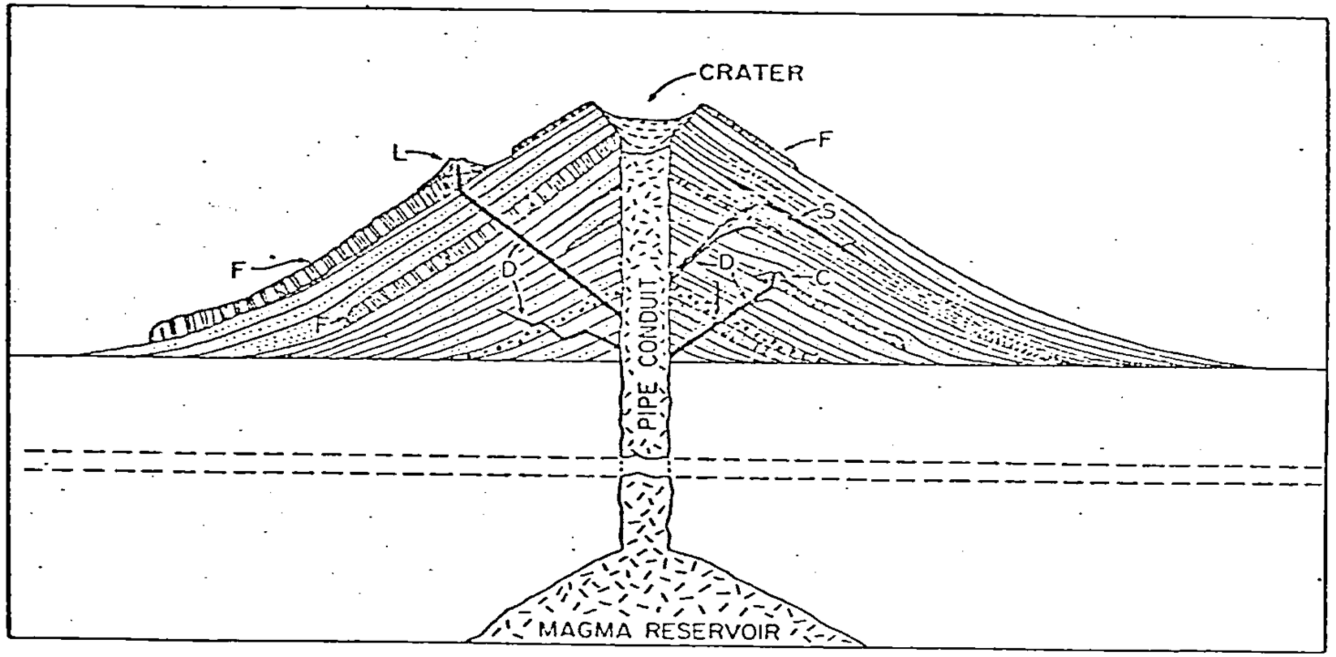


FIGURE 2-1. Diagram of a typical composite volcano, showing the cone, crater, central pipe conduit, magma reservoir, dikes (D), a dike conduit feeding a lateral cone (L) and lava flow (F), a buried cinder cone (C), and a sill (S). Tephra layers are dotted, breccia layers are marked with small triangles, and lava flows (F) are irregularly cross-hatched.

Estructuras

Estructuras

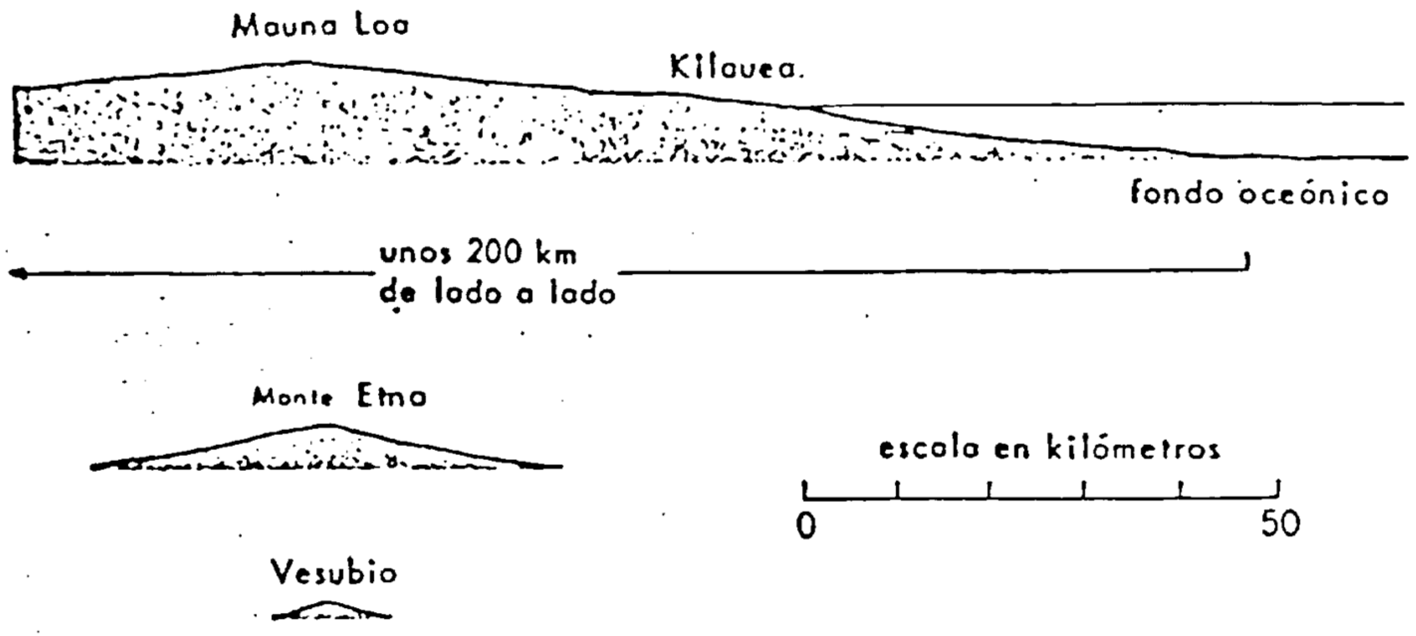
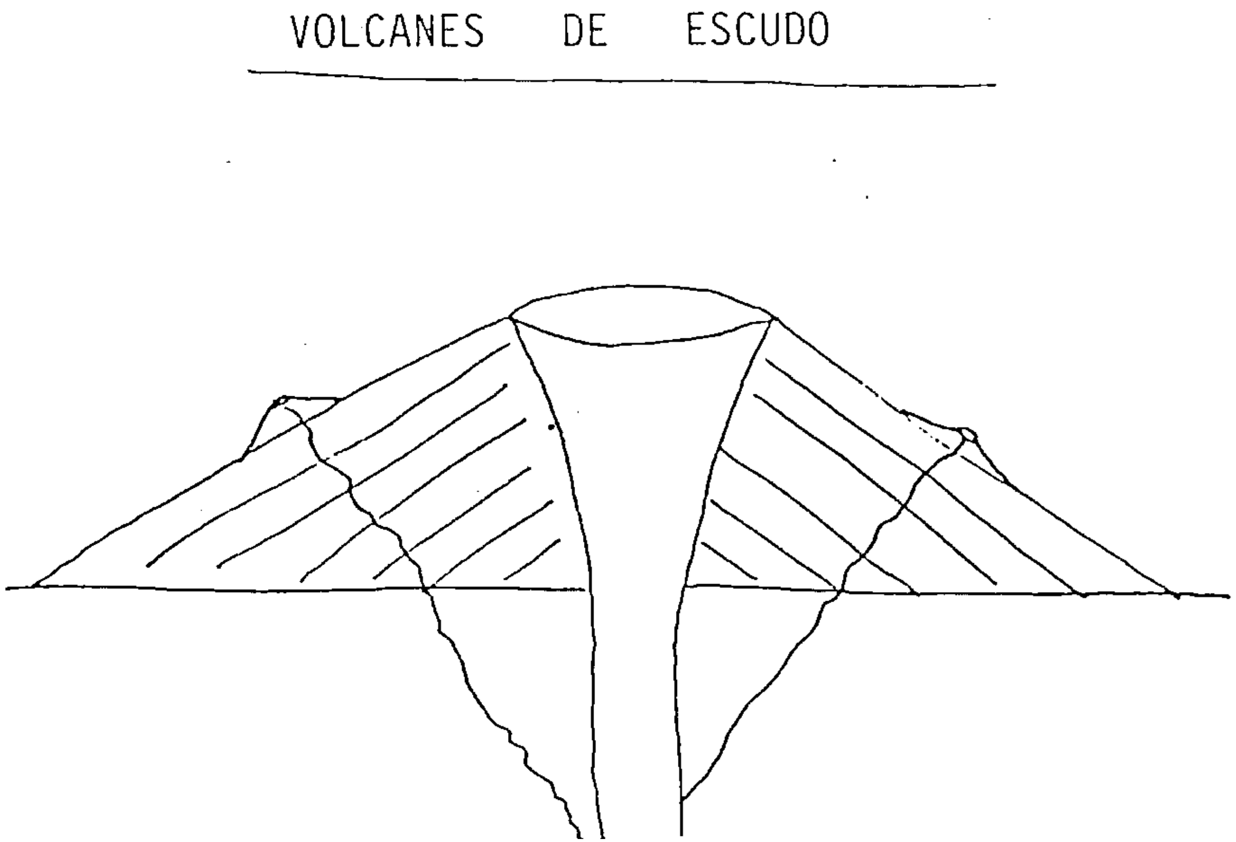


Fig. 5-1. Tamaño de algunos volcanes. (Modificado a partir de Menard, 1964.)

Estructuras

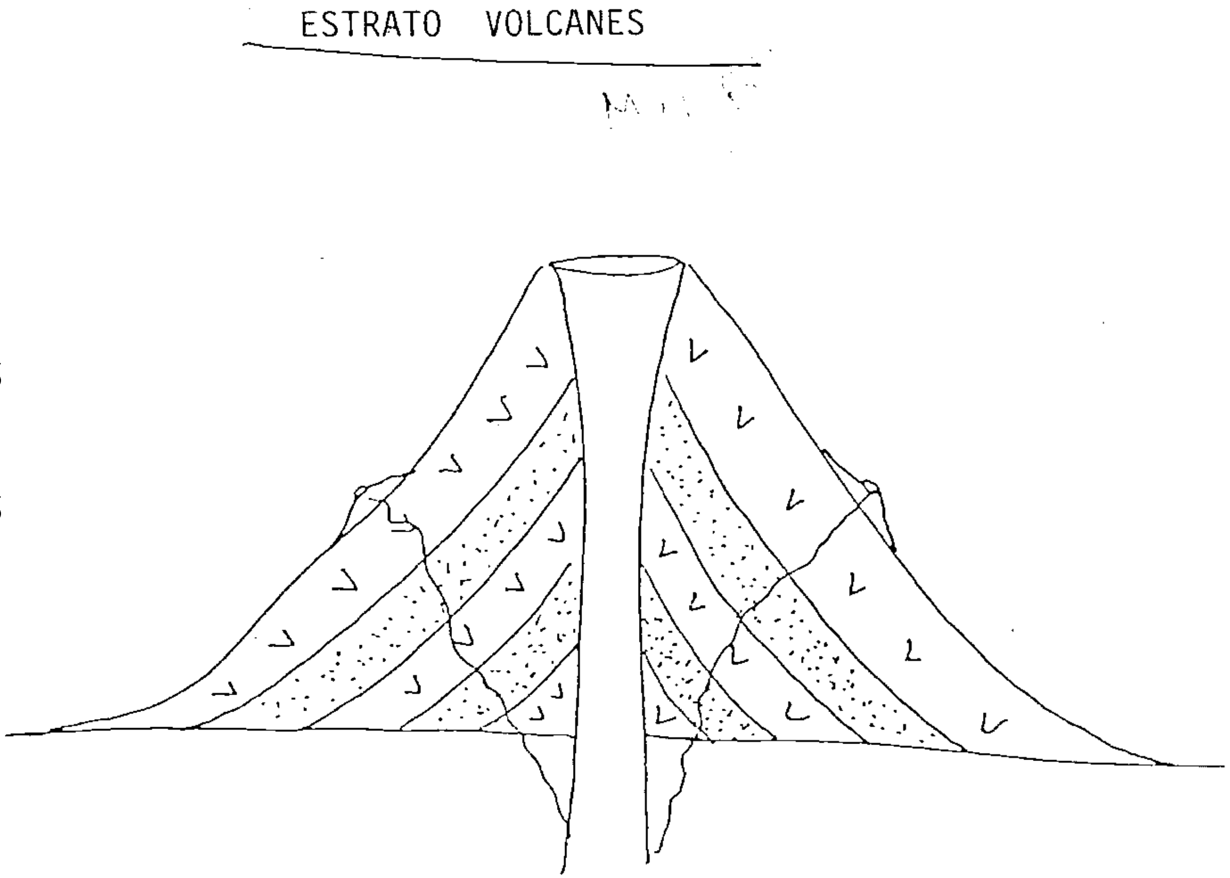
Estructuras

- Lava
- Conos amplios
- Poca pendiente
- Salidas fisurales
- Hawái, Xitle



Estructuras

- Lava y piroclásticos
- Salidas principales
- Fracturas y/o conos parásitos
- Grandes dimensiones y gran pendiente
- Popocatépetl, Fujiyama



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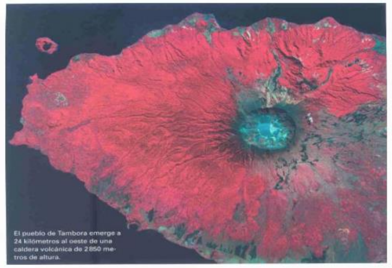
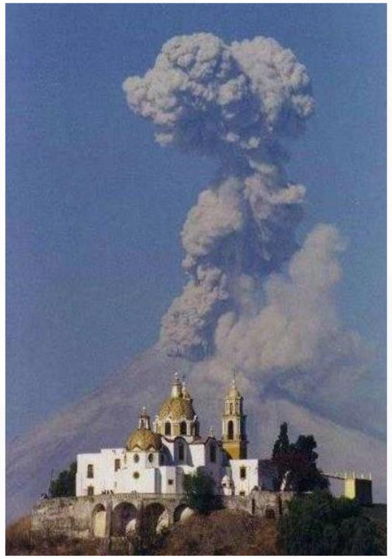
Estructuras





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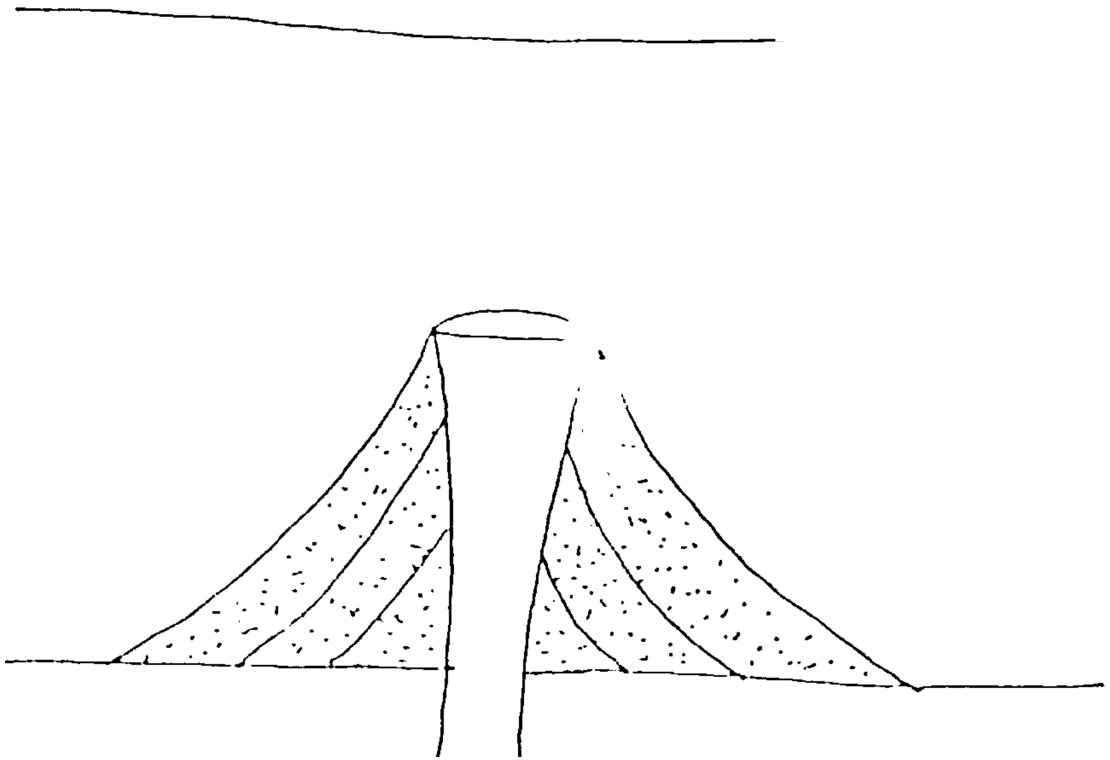
Estructuras

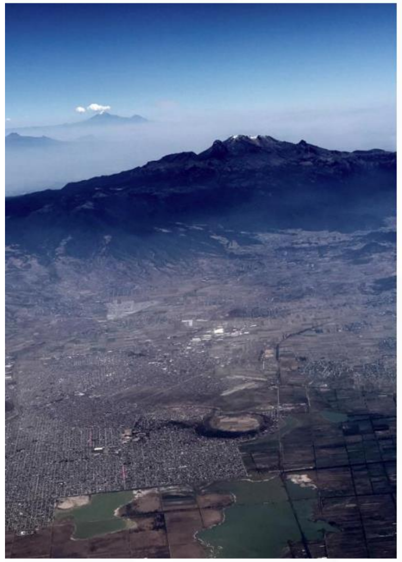
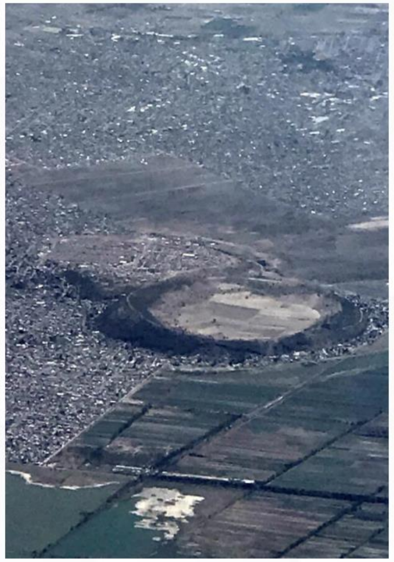
- Material piroclástico
- Edificios pequeños
- Pendientes fuertes

CONOS CINERITICOS

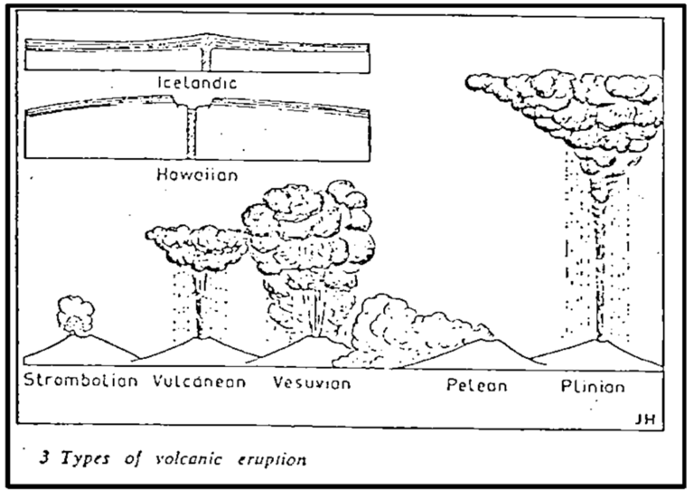


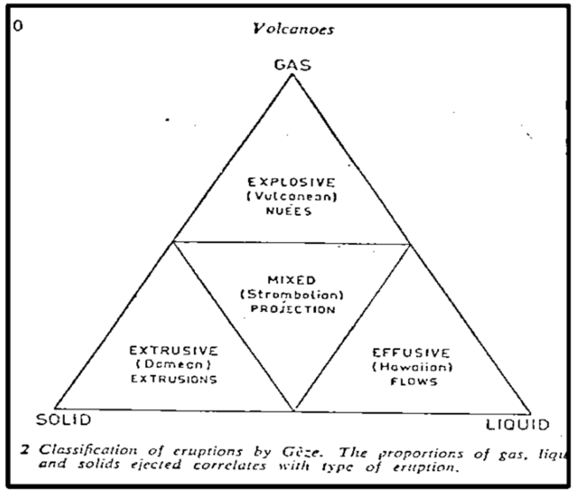
Estructuras

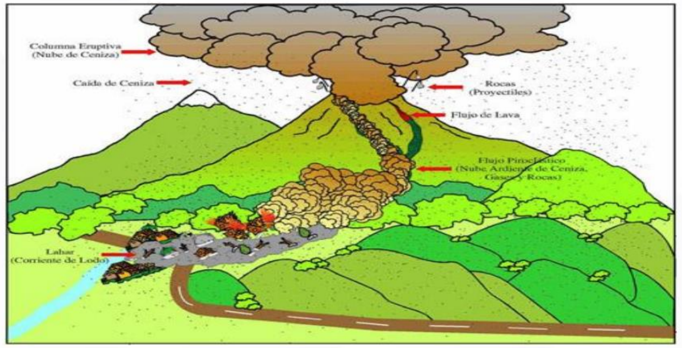


Estructuras

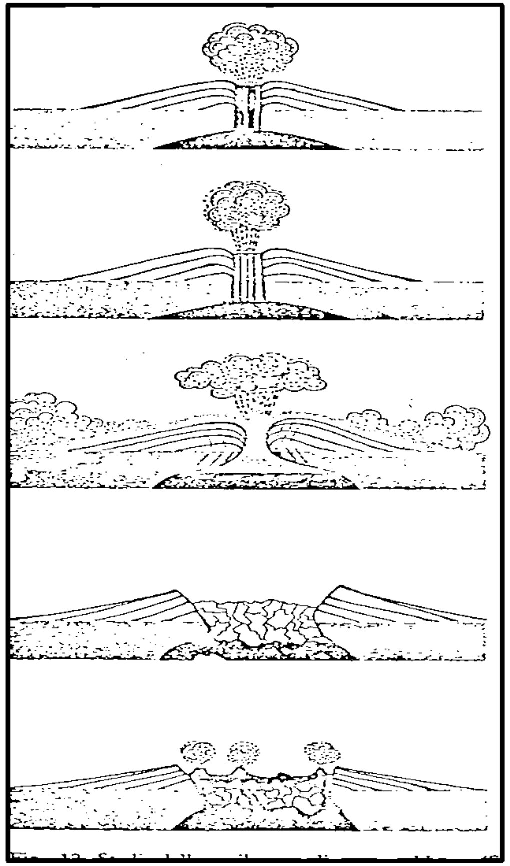






Estructuras

Estructuras



06 4.TECAGREG Tecnologia de agregados

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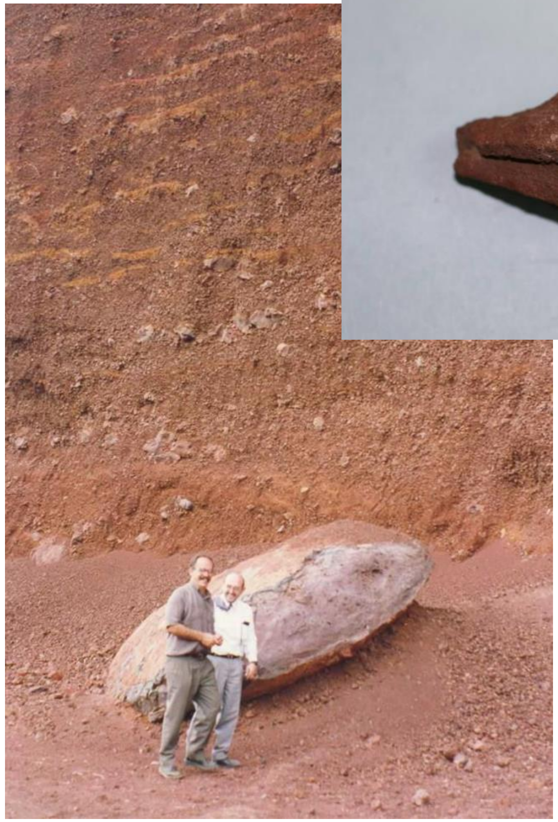
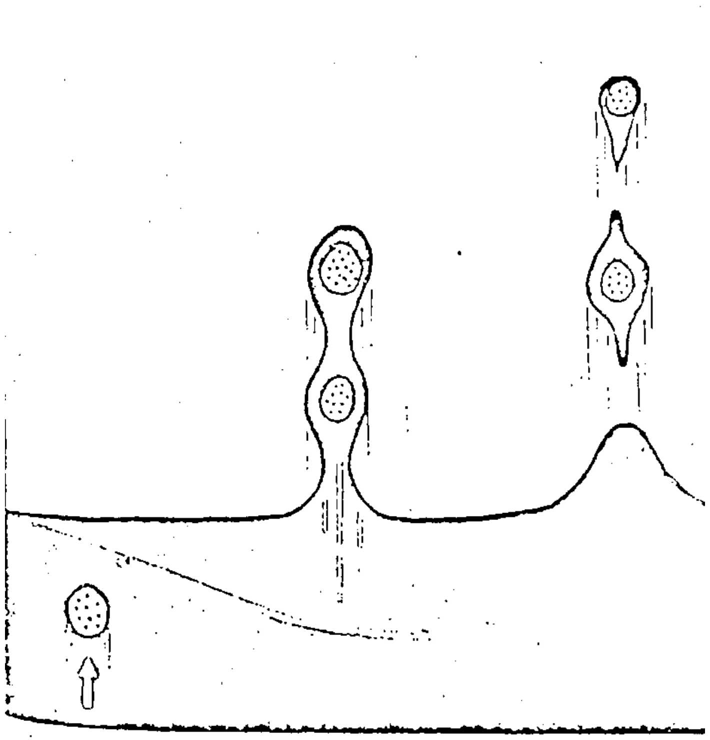
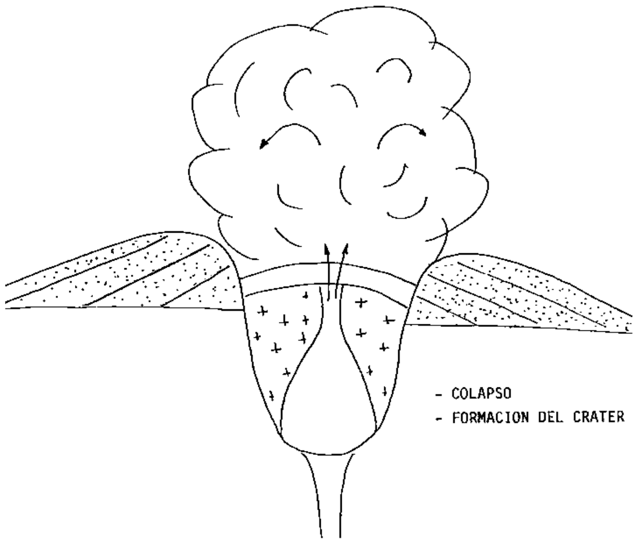
Estructuras



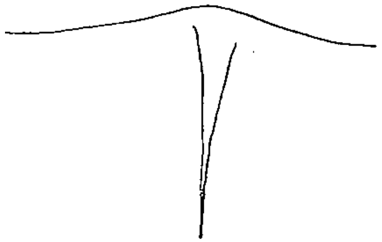


22 *Formation of volcanic bombs from a ribbon of lava with pr inclusions. The terminal bomb is called uni-polar, the twc bombs are bi-polar.*

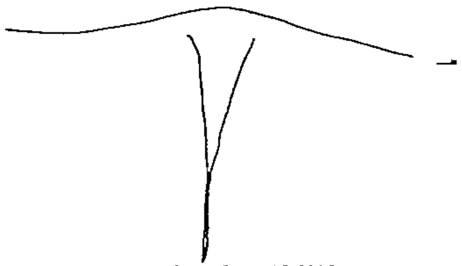
Estructuras
Maar o Xalapascos



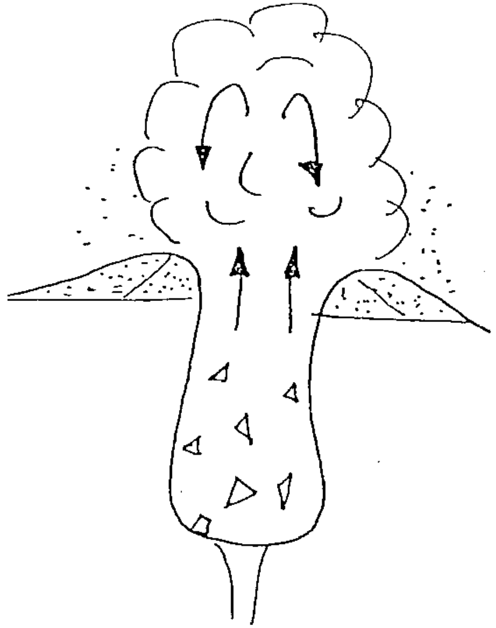
- COLAPSO
- FORMACION DEL CRATER



- FRACTURAMIENTO



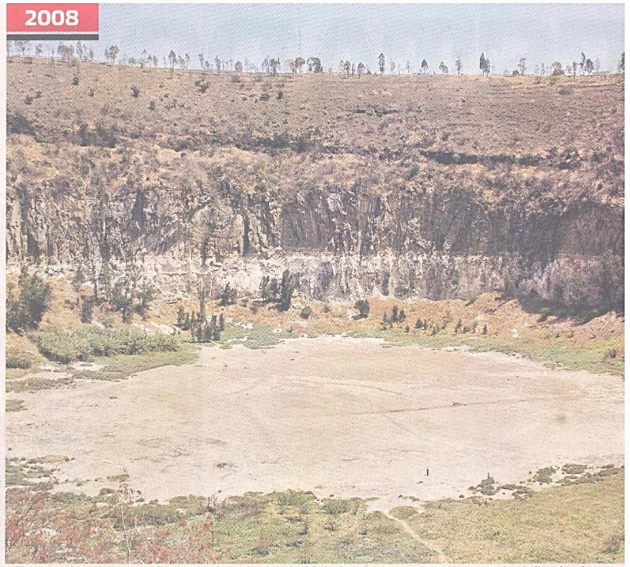
- ASCENSO DEL MAGMA



- EXPLOSIONES FREATOMAGMATICAS
- MATERIAL JUVENIL Y PREXISTENTE

Estructuras

Al lago que estaba en el cráter de un volcán en Valle de Santiago, le llamaban La Alberca, y ahí iban personas a nadar.

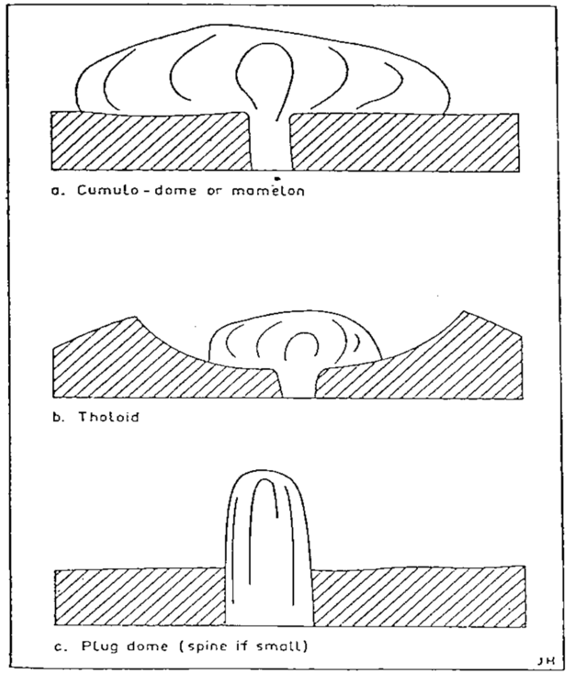


El cráter, ahora seco, tiene 70 metros de profundidad y 90 de diámetro; en sus paredes se ve donde llegaba el agua.

Estructuras

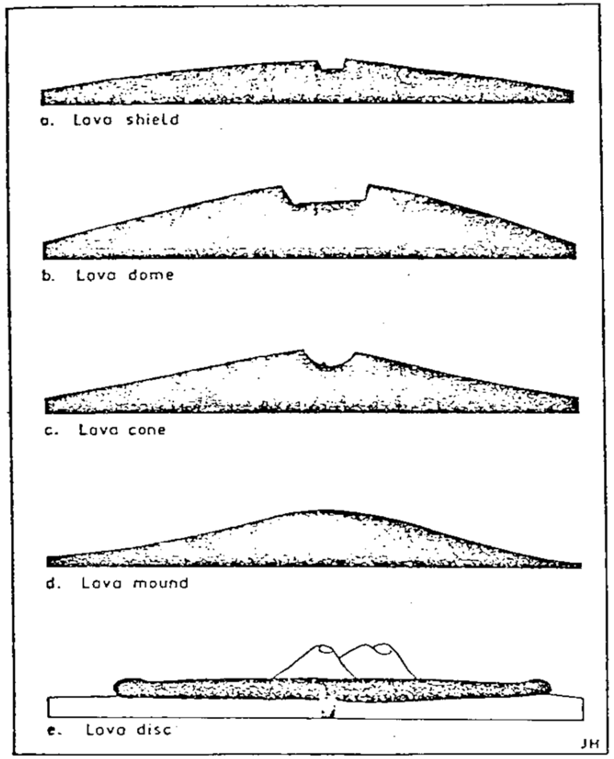
ACID LAVA VOLCANOES

Acid igneous rocks are generally very viscous, and if they do not explode their lack of flow gives rise to a number of distinctive landforms (Fig. 5).



5 Diagrammatic representation of acid lava landforms (not to scale)

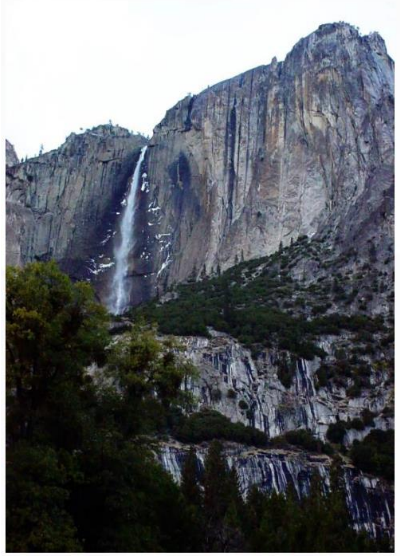
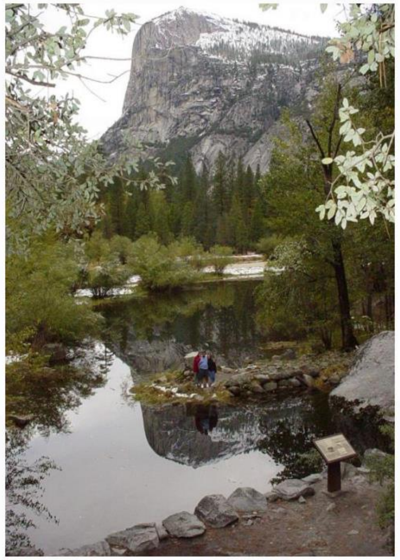
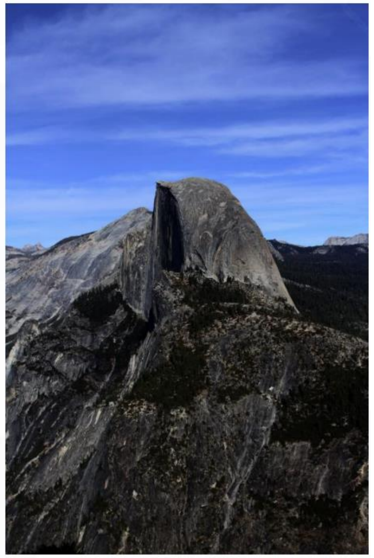
Types of Volcano 21



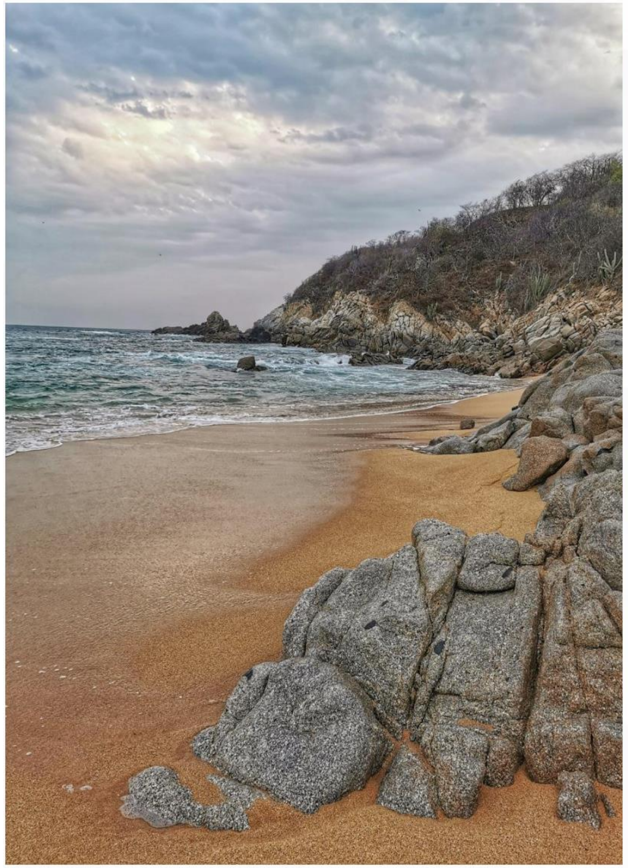
4 Diagrammatic representation of basaltic volcano types (not to scale)

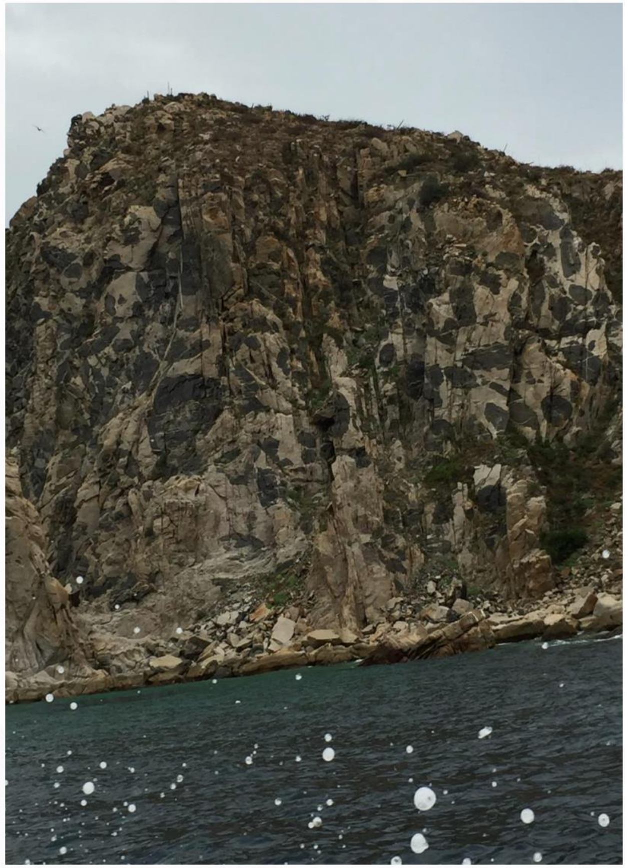
Estructuras



Estructuras





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Estructuras

Estructuras









04

Rocas ígneas

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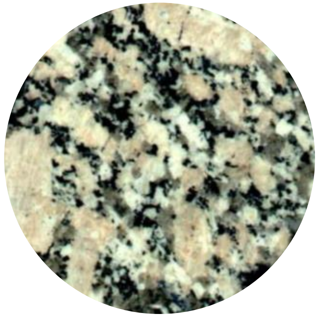
Rocas ígneas

Rocas intrusivas

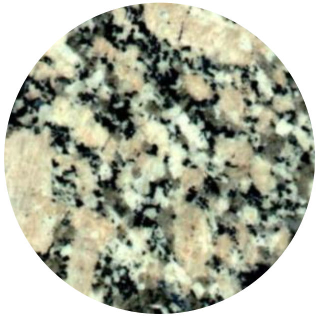
- Plutónicas



Rocas ígneas

Rocas intrusivas

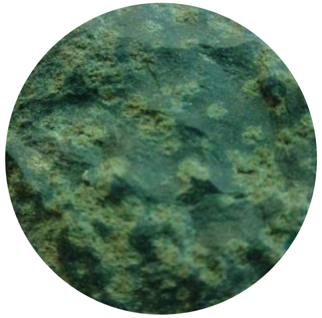
- Plutónicas
- Hipabisales

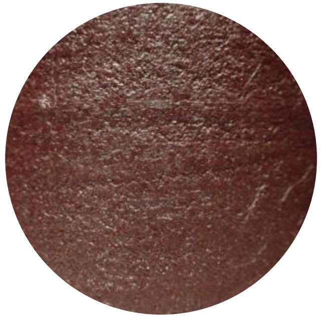





Rocas extrusivas

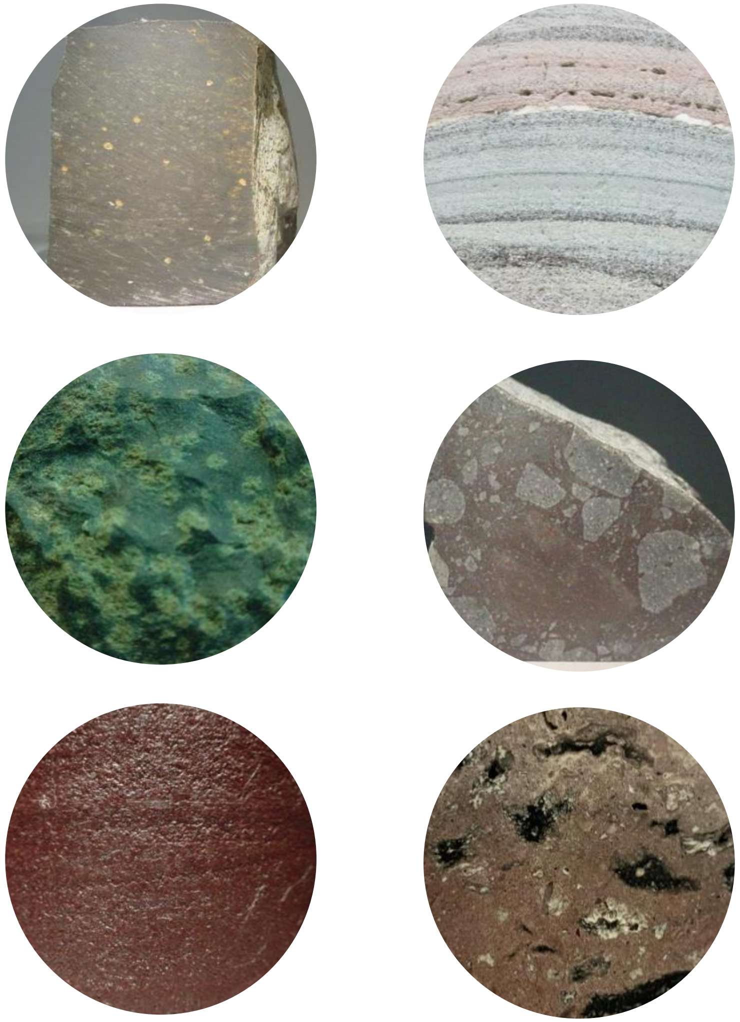
- Volcánicas





Rocas extrusivas

- Volcánicas
- Piroclásticas



Rocas ígneas
Clasificación

Rocas ígneas
Clasificación en base al color

Johansen

Ácidas.................. 5%

Intermedias............50%

Básicas..................95%

Shand

Leucocraticas...........Maficos /< 30%

Mesocraticas.............Maficos entre 30 y 60%

Melanocraticas..........Maficos entre 60 y 90%

Hipermecanicas.........Maficos > 90%

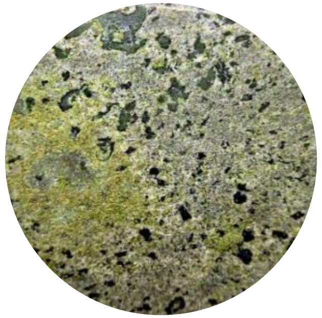
Ellins

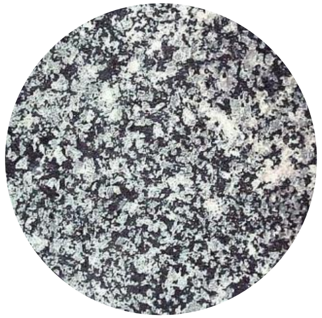
Holofelsicas............./ < 10% de Maficos

Felsicas.................... entre 10 y 40%

Mafelsicas..................entre 40 y 70%

Maficas.....................> 70%





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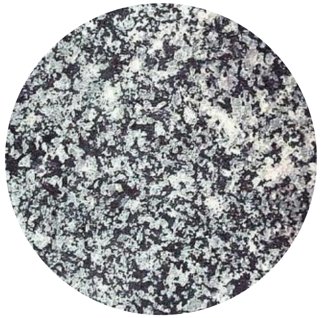
Rocas ígneas

Composición química.	Alto cont. de cuarzo		Alto cont. de Hierro y Magnesio		
Color	Predominan minerales claros		Predominan Minerales oscuros		TEXTURA
grano grueso	GRANITO	DIORITA	GABRO	PERIDOTITA	FANERITICA
fino	RIOLITA	ANDESITA	BASALTO	(muy rara)	AFANITICA
Composición Mineral.	Cuarzo Feldespatos Ferromagnesianos				
Composición	Alto cont. de sílice Principalmente vidrio				
Vítrea	Obsidiana y Pómez		-	-	VITREA
Fragmentada	Toba y Brecha volcánica			-	PIROCLASTICA



Rocas ígneas

Composición química.	Alto cont. de cuarzo			Alto cont. de Hierro y Magnesio	
Color	Predominan minerales claros			Predominan Minerales oscuros	TEXTURA
grano grueso	GRANITO	DIORITA	GABRO	PERIDOTITA	FANERITICA
fino	RIOLITA	ANDESITA	BASALTO	(muy rara)	AFANITICA
Composición Mineral.	Cuarzo Feldespatos Ferromagnesianos				
Composición	Alto cont. de sílice Principalmente vidrio				
Vítrea	Obsidiana y Pómez	-		-	VITREA
Fragmentada	Toba y Brecha volcánica			-	PIROCLASTICA

Rocas ígneas

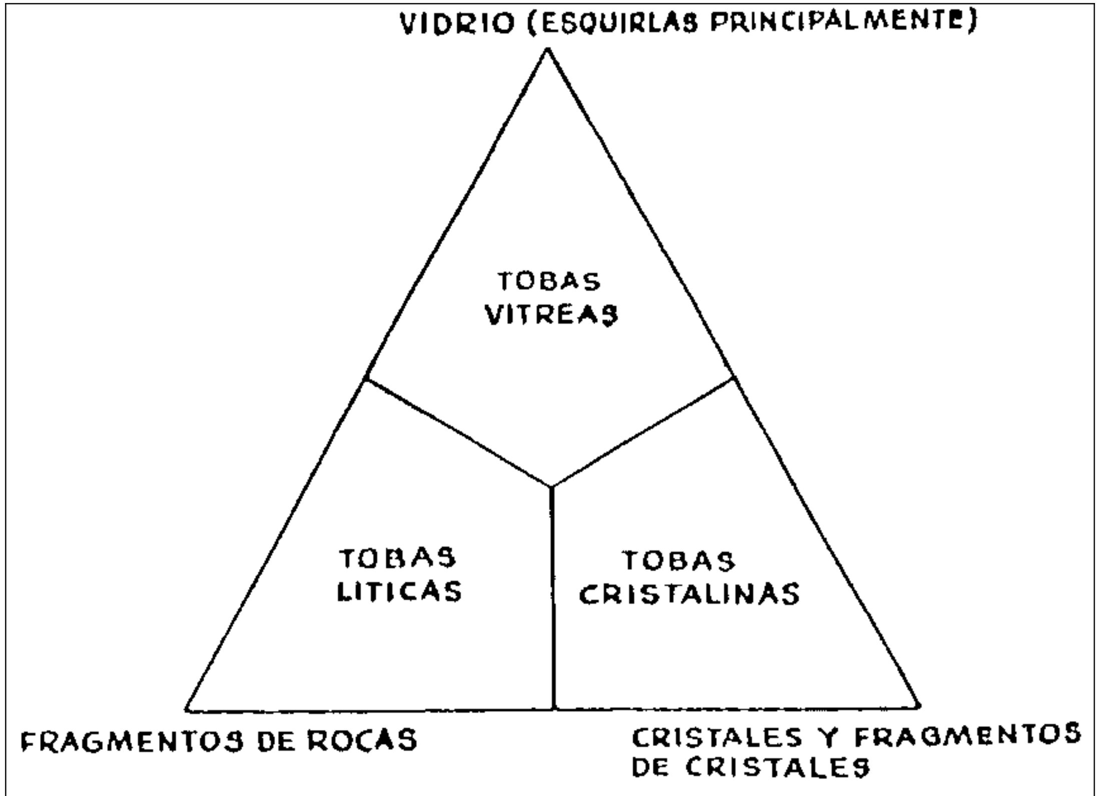
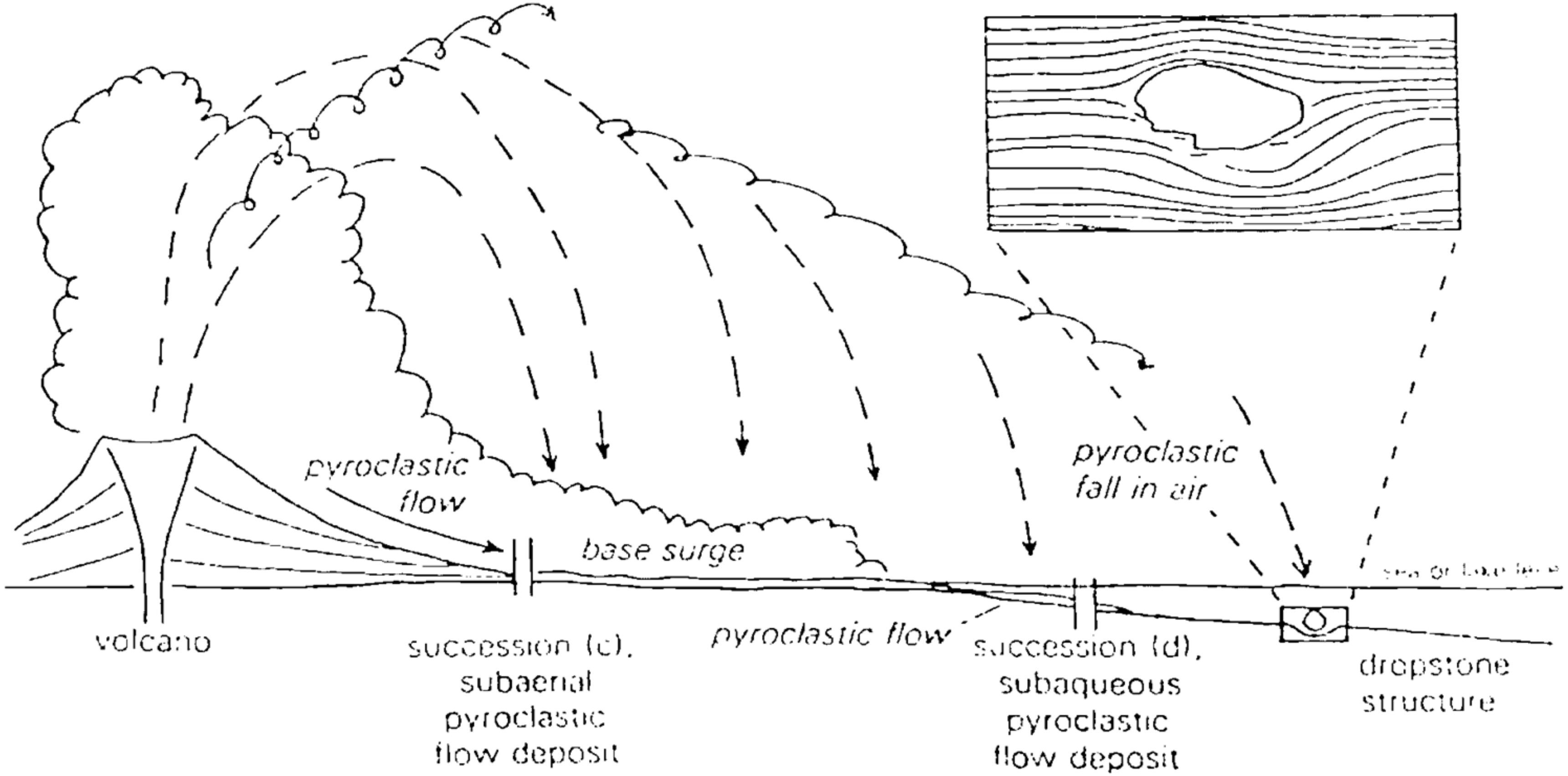
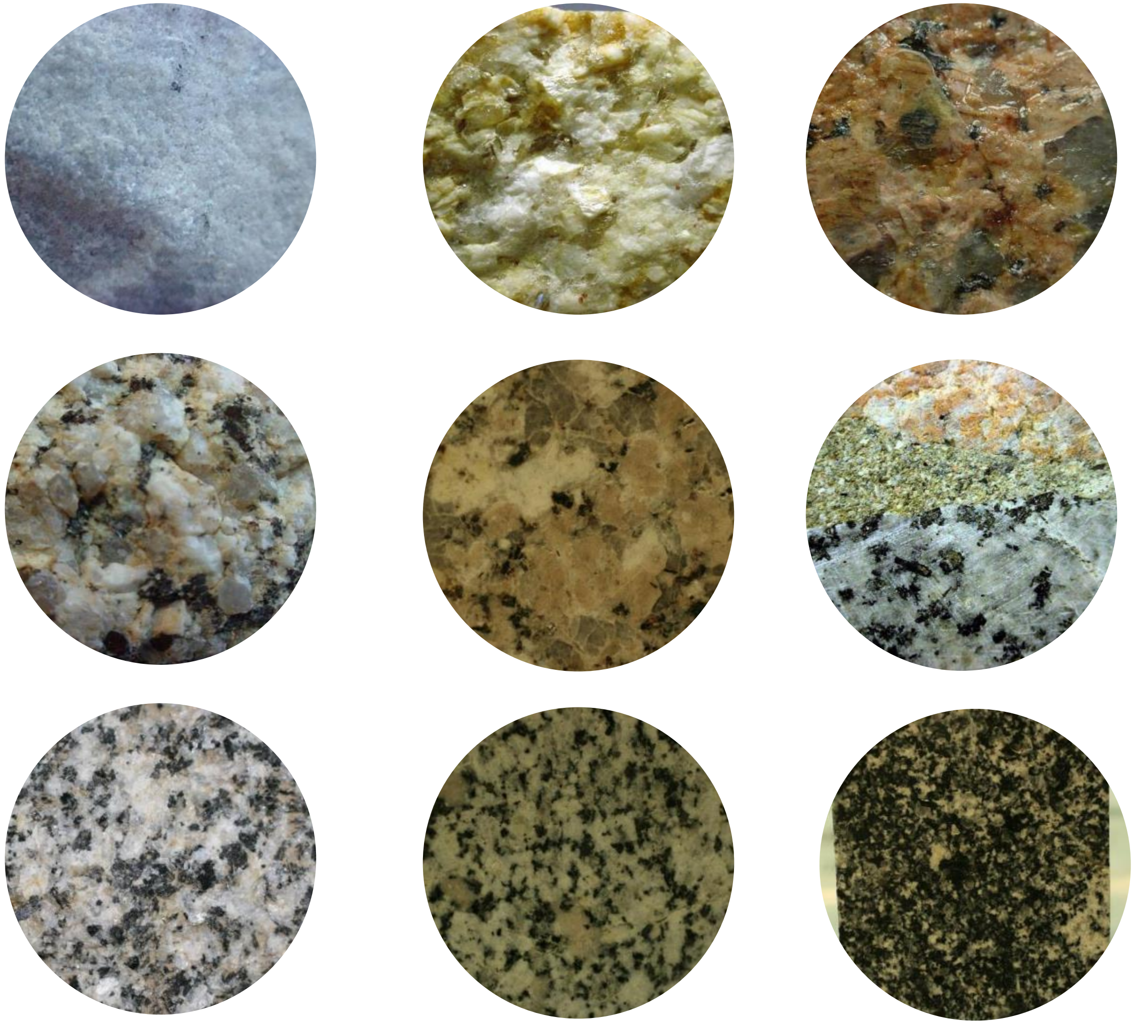


FIG. 24. — Clasificación de las tobas.

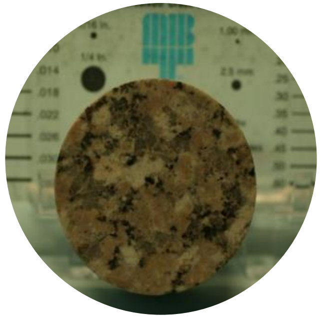
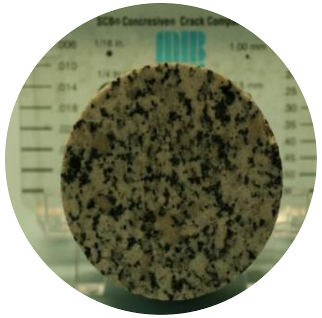
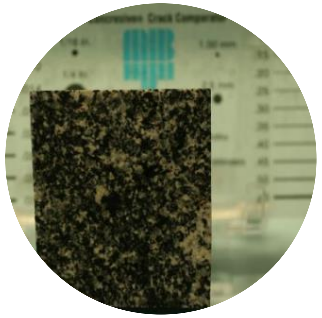
Rocas ígneas

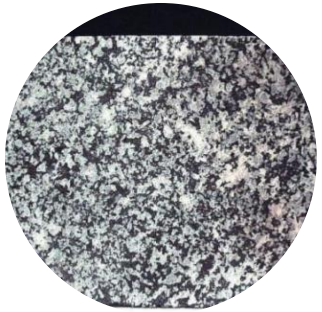


Rocas ígneas
Índice de color

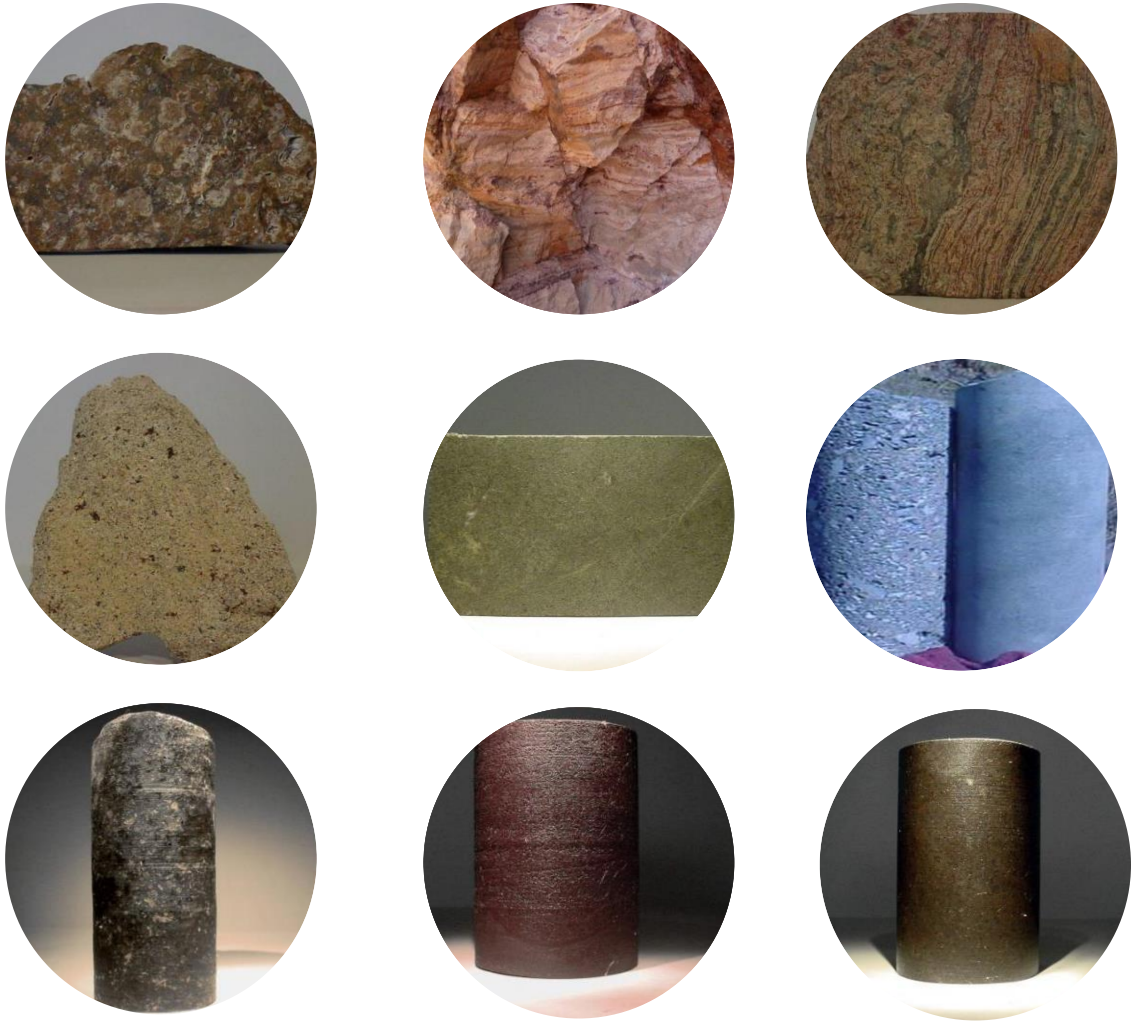


Rocas ígneas
Intrusivas

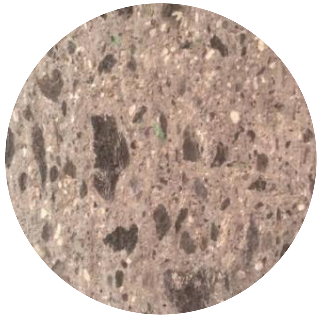




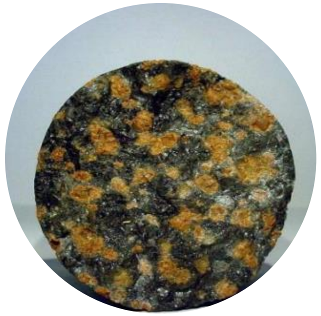
Rocas ígneas
Extrusivas

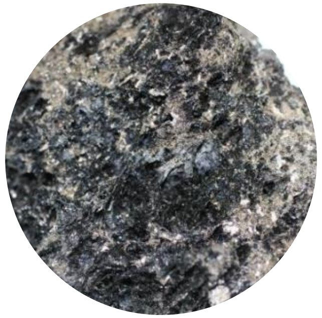


Rocas ígneas
Extrusivas/Piroclásticas





Rocas ígneas

Extrusivas/Piroclásticas









Gracias

Por su participación

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